

Incident Report Analysis: DoS Attack via ICMP Flood

Analysis applying the NIST Cybersecurity Framework (CSF), completed as part of the Google Cybersecurity Professional Certificate

Summary

The organization experienced a distributed denial-of-service (DoS) attack that disrupted the internal network for two hours. The attack originated from a flood of ICMP packets that entered the network through an unconfigured firewall, overwhelming network services and preventing normal internal traffic from accessing network resources. The incident management team responded by blocking incoming ICMP packets, taking non-critical network services offline, and restoring critical services. A subsequent investigation by the cybersecurity team confirmed that the unconfigured firewall was the entry point that allowed the malicious actor to flood the network.

Identify

Asset and technology impact: The attack traced back to the organization's network firewall, which had no rules configured to filter or rate-limit incoming ICMP traffic. This gap allowed the flood of ICMP packets to reach the internal network directly, overwhelming network services and blocking legitimate internal traffic for two hours.

Business impact: Any business process depending on internal network resources, including internal applications and communication tools, was disrupted for the duration of the outage, since normal traffic could not reach network services.

Access: Network and security administrators require ongoing access to firewall configuration settings and network monitoring systems to maintain and audit these controls going forward.

Protect

Access control: Firewall rules should limit the rate of incoming ICMP packets and verify source IP addresses, blocking spoofed or unauthorized traffic from reaching internal systems.

Protective technology: An intrusion prevention system (IPS) should be deployed to filter ICMP traffic exhibiting suspicious characteristics before it reaches the internal network.

Maintenance: Firewall configurations should be reviewed on a scheduled basis to confirm that filtering rules remain in place and current, since the root cause of this incident was a firewall left unconfigured rather than a bypass of existing controls.

Training: Network and security staff should be trained on proper firewall configuration and maintenance procedures to prevent similar gaps from going unnoticed.

Detect

Continuous monitoring: Network monitoring software should track traffic patterns in real time and flag sudden spikes in ICMP volume or other anomalies consistent with a denial-of-service attempt.

Detection process: An intrusion detection system (IDS), paired with a security information and event management (SIEM) tool, should be used to correlate alerts and notify the security team as soon as suspicious traffic patterns are identified.

Audit cadence: Firewall logs should be reviewed on a regular, ongoing basis, not only in response to an incident, to catch configuration gaps before they can be exploited.

Respond

Mitigation: The response plan should call for immediately blocking the offending traffic type at the firewall and taking non-critical services offline to preserve capacity for critical systems, consistent with the steps taken in this incident.

Analysis: Firewall and network logs should be reviewed to trace the attack's entry point and confirm the scope of systems affected.

Communications: The incident and its resolution should be communicated promptly to IT staff and affected internal stakeholders so that downstream teams understand the disruption and its cause.

Improvements: Following resolution, the response process should include a review step to confirm what allowed the incident to occur, in this case an unconfigured firewall, so the same gap is addressed rather than only the immediate symptom.

Recover

Recovery planning: The organization should maintain current firewall configuration backups and network topology documentation to restore services quickly and correctly once a threat is contained.

Improvements: Newly implemented protections, including ICMP rate limiting, source IP verification, and IPS/IDS coverage, should be verified as functioning correctly before the incident is considered fully closed.

Communications: Restoration status should be communicated to relevant internal stakeholders throughout the recovery process, so that affected teams know when normal operations have resumed.

Recommendations & Next Steps

The root cause of this incident was a firewall left without ICMP filtering rules, not a sophisticated attack technique. This makes the fix straightforward and the priority order clear:

- 1. Immediate priority:** Confirm ICMP rate limiting and source IP verification are active on the firewall. These are low-cost, high-impact controls that directly close the gap exploited in this incident and should already be in place before any other measure is pursued.
- 2. Near-term priority:** Establish a recurring firewall configuration review (e.g., monthly, or triggered by any abnormal traffic alert) so that a similar misconfiguration cannot persist undetected in the future.
- 3. Longer-term investment:** Evaluate and deploy IDS/IPS and SIEM capabilities if not already in place. These require more budget and setup time than the firewall fixes above, but provide ongoing detection value beyond this specific attack type.

Reflection: This incident is a reminder that foundational configuration hygiene, not just advanced tooling, is often the deciding factor in whether an attack succeeds. Applying the NIST CSF here made it easier to

separate the immediate fix (Protect) from the longer-term monitoring investment (Detect), rather than treating the response as a single undifferentiated action item.